StabilitySection: Cross-Census Stability of GeoSection-Optimal Redistricting Maps

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Abstract

StabilitySection is not a new map constructor. It is a cross-census audit framework for asking whether GeoSection selects the same first bisection, similar district assignments, and similar outcome diagnostics when the input census year changes. We define the **Census Stability Score** (CSS), a composite metric for recording this cross-census robustness, and separate three levels of evidence: implemented GeoSection runs, tract-split proxy comparisons, and future district-level assignment matching.

The paper's central claim is methodological rather than legal-dispositive. When the same geography-first procedure returns the same first-level structure under multiple census snapshots, that is evidence that the structure is not an artefact of one vintage of population data. It does not by itself prove that a map is legally required, partisan-neutral, or immune from challenge. It gives courts, commissions, and researchers an auditable way to distinguish stable geographic structure from census-vintage sensitivity.

We present the 2020 GeoSection baseline, a 2000–2010 tract-split proxy comparison for 30 comparable state cases, and the CSS computation plan for the full 2000/2010/2020 package. The current empirical result is deliberately narrow: under a 5 percentage point first-split threshold, 20 of 30 comparable cases are stable in the proxy table. Full three-census CSS, district Jaccard matching, and final-plan partisan comparisons remain Phase 2 evidence.

1 Introduction

GeoSection [Della-Libera, 2026b] introduced isoperimetrically-normalised ratio-optimal bisection for congressional redistricting, demonstrating that minimum-edge-cut recursive bisection with a $\sqrt{\min(i, k-i)}$ normalisation selects a *natural ratio* for each state that reflects genuine geographic structure rather than computational artefact. B.7 [Della-Libera, 2026a] established that this natural ratio is *seed-stable* within a single census year: running GeoSection with 1000 different random seeds produces the same natural ratio in $> 95\,\%$ of runs for most states.

This paper asks a deeper question: is the natural ratio *census-stable*?

The census-stability question. Every ten years, the Census Bureau publishes new population data, tract boundary updates, and a new congressional apportionment. Redistricting commissions and courts must draw new district maps from scratch. The question for an algorithmic redistricting system is: if we run GeoSection on 2000 Census data, then on 2010 data, then on 2020 data, do we get the same answer?

If yes, the result is meaningful evidence. The 2000 census measured one population distribution. The 2010 census measured a different distribution—ten years of births, deaths, migration, and urban development later. The 2020 census measured yet another. Despite these three independent snapshots of American demographic reality, the algorithm converges to the same geographic partition. That partition is not a product of any particular census cycle. It is evidence for a persistent *geographic structure of the state*.

Why this matters legally. One useful defence against a partisan gerrymandering claim is a record showing that a challenged map follows geography-first rules rather than a desired partisan outcome. B.7's seed stability provides one dimension of this defence: the map is the same regardless of which random seed initialised the algorithm. Census stability provides a second, stronger dimension.

Consider a state where GeoSection has produced the same 5 D/9 R seat distribution in 2000, 2010, and 2020. A challenger who claims the map was drawn only to achieve Republican advantage must then confront a measurable counterfactual: the algorithm produced the same answer under three different population datasets spanning twenty years. The -13 pp proportionality gap is not a product of 2020's specific partisan distribution. It was the result in 2000, when Republicans had a different geographic distribution. It was the result in 2010, after Obama-era demographic shifts. It was the result in 2020. The geographic sorting of Democratic voters into concentrated urban areas is the cause of the gap—and that sorting is stable across census cycles.

In state-court litigation under post-*Rucho* doctrine, a census-stable GeoSection map can support a geographic explanation record. It cannot replace state constitutional law, a comparison to enacted maps, or jurisdiction-specific facts, but it can make the geography question falsifiable.

The Census Stability Score. We formalise this intuition as the **Census Stability Score** (CSS), a composite metric computed for each state by comparing GeoSection outputs across census years:

$$\mathrm{CSS(state)} = 0.5 \cdot s_{\mathrm{seat}} + 0.3 \cdot s_{\mathrm{ratio}} + 0.2 \cdot s_{\mathrm{gap}},$$

where s_{seat} measures partisan-outcome stability, s_{ratio} measures natural-ratio stability, and s_{gap} measures proportionality-gap-magnitude stability. A CSS of 1.0 means identical seat counts, identical natural ratios, and identical proportionality gaps across all census years. CSS ≥ 0.90 is the threshold for "highly stable": the state's geographic structure is durably expressed by GeoSection across decades.

What StabilitySection actually sees. The first split is the clearest place to visualize the method. Suppose a four-seat state is tested in two census years. StabilitySection does not merely ask whether both years say “2:2”; it records the block split that made that ratio concrete:

2000 candidate	A	A	B	B	$f_{2000} = 4/8 = 0.50$
	A	A	B	B	
2010 candidate	A	B	B	B	$f_{2010} = 4/8 = 0.50$
	A	A	A	B	

The ratio is stable in both rows, but the district-assignment picture may still move. That is why the paper distinguishes first-split ratio stability, first-split tract-share stability, and the later district-level Jaccard match.

Two sources of change. A map may change between census years for two reasons. *Type I*: population shifts within a fixed geographic topology. People move between tracts, altering population balance targets, but the underlying tract graph is unchanged. This is a smooth perturbation of the METIS input weights. *Type II*: tract boundary redesign. The Census Bureau redraws some tract boundaries between decennial censuses, splitting or merging tracts. This changes the graph topology and can propagate to downstream district boundaries. StabilitySection separates these two effects, measuring each independently.

Relationship to B.7 and T.1. B.7 measures stability along the *seed dimension*: fixed census year, varying initialisation. T.8 measures stability along the *census dimension*: fixed algorithm, varying input data. Together they provide a 2D stability picture: $(s_{\text{seed}}, s_{\text{census}})$. States with high stability in both dimensions have a strong geographic determinism claim. States with low stability in both dimensions require the most careful procedural justification for any particular map they produce.

Our contributions.

1. **The Census Stability Score (CSS)** (§3): a formally defined composite metric for cross-census algorithmic stability, with sub-components measuring seat, ratio, and gap stability separately.

2. **Lorenz drift as a predictive proxy** (§3.4): the shift in the optimal ratio p^* on the population-geography Lorenz curve between census years predicts CSS; states where p^* drifts significantly are structurally volatile.

3. **Two-source decomposition** (§3.5): population-only perturbation stability vs. full cross-year stability, isolating the contribution of tract-boundary redesign.

4. **50-state 2020 GeoSection baseline** (§4): complete natural ratio and proportionality results for 44 multi-district states, providing the 2020 anchor for the cross-census comparison.

5. **Legal and policy implications** (§5.1): the CSS as an evidentiary record for commissions, courts, and proposed statute design, with legal conclusions left to case-specific analysis.

Data status. The 2020 GeoSection baseline is the anchor table in this paper. The 2000–2010 tract-split proxy table covers 30 comparable state cases. Full CSS still requires archived year-by-year GeoSection outputs, a common geographic crosswalk for district Jaccard matching, and final-plan outcome analysis for the same plan packages.

Paper organisation. Section 2 reviews related work. Section 3 defines StabilitySection and the CSS formally. Section 4 presents the 2020 baseline and available cross-census comparison. Section 5 discusses legal implications and predictive modelling. Section 6 concludes.

2 Related Work

2.1 Within-Year Seed Stability: B.7

The companion paper B.7 [Della-Libera, 2026a] establishes seed stability for minimum-edge-cut redistricting within a single census year. Running 1000 different random initialisations of the METIS algorithm on 2020 Census data, B.7 finds that the natural ratio is consistent across seeds for $> 95\,\%$ of states and that partisan seat counts vary by at most one seat in competitive states. This establishes the MEC solution space as constrained: the algorithm converges to a small number of topologically distinct solutions, with the minimum-edge-cut solution dominating.

T.8 extends this stability analysis along a second dimension. Where B.7 fixes the input data and varies the seed, T.8 fixes the algorithm and varies the input data across three independent decennial surveys. The two stability dimensions are complementary: a state with high seed stability but low census stability has an algorithm that reliably finds its natural partition given any one census snapshot, but whose natural partition changes as demographics evolve. A state with high census stability but low seed stability would have a durable geographic structure that is difficult to reliably detect. In practice, we expect seed stability and census stability to be correlated: states with simple geographic structure (clear urban cores) are both easy to detect (high seed stability) and geographically durable (high census stability).

2.2 GeoSection and Isoperimetric Normalisation: T.1

GeoSection [Della-Libera, 2026b] introduced the normalised edge-cut $\mathrm{EC}/\sqrt{\min(i, k-i)}$ as the criterion for selecting the natural first-level split ratio. The key finding is that for states with compact, geographically isolated metropolitan cores, GeoSection *confirms* the asymmetric split that unnormalised minimum-edge-cut would also select, while for states with distributed population, it correctly selects a more balanced split.

For cross-census stability, GeoSection's normalisation provides a useful stability argument: the denominator $\sqrt{\min(i, k-i)}$ is a function of the *number of districts*, not the population distribution. If the state's seat count is unchanged between censuses (the common case), the normalisation formula is identical; only the edge-cut numerator changes with population. We conjecture in §1 that the natural ratio is stable whenever the population shift is small relative to the isoperimetric gap between the winning and second-best ratios.

2.3 Temporal Stability of Redistricting Maps: Literature

Temporal stability of redistricting outcomes has received limited direct treatment in the quantitative redistricting literature, but several threads are relevant.

Population geography as a stable driver. Rodden [2019] documents the deep roots of the urban-rural political divide in American geography. The geographic sorting of Democratic voters into dense urban areas is not a recent phenomenon: it has been building since at least the 1970s, accelerating through the 1980s and 1990s, and is now a structural feature of American political geography. Chen and Rodden [2013]'s analysis of "unintentional gerrymandering" shows that compact, geography-only redistricting algorithms systematically produce Republican-advantaged maps because of this geographic sorting—and that this bias has been present across multiple redistricting cycles. Both findings predict high cross-census stability for GeoSection: if the underlying geographic structure is stable, the algorithm's output should be stable.

MCMC ensemble stability. Chikina et al. [2017] and DeFord et al. [2019] (GerryChain/ReCom) use Markov chain Monte Carlo methods to sample from the space of valid redistricting plans and assess whether an enacted plan is an outlier. These methods have been applied primarily within a single census year; their cross-census behaviour is not characterised. StabilitySection provides a complementary analysis: instead of sampling the space of plans within one year and measuring variance, we hold the algorithm fixed and measure variance across years.

Partisan stability in the literature. McGhee [2014] and Stephanopoulos and McGhee [2015] introduced the efficiency gap as a measure of partisan bias in redistricting. Their analysis of historical congressional maps shows that efficiency gaps are often stable across redistricting cycles within a state—a form of empirical cross-cycle stability driven by population geography. Our CSS provides a more fine-grained algorithmic measure of the same underlying phenomenon.

2.4 Census Data Changes Between Decennial Years

Between each pair of consecutive decennial censuses, the Census Bureau makes two types of changes relevant to redistricting:

Population counts. The total population and its geographic distribution change. Between 2000 and 2020, the US population grew by 18.2 %, from 281.4M to 331.4M. Growth was highly uneven: the Sun Belt states (TX, AZ, GA, FL, NV) grew by 30–60 %; declining states (WV, MS, AL) contracted. Population shifts within states are even larger: suburban expansion around major metros, rural depopulation, and the concentration of population in a smaller number of very large metro areas.

Tract boundary redesign. Between 2000 and 2010, the Census Bureau split, merged, and redrawn approximately 15 % of census tracts. A similar rate of redesign occurred between 2010 and 2020. The adjacency graph G used by GeoSection is therefore not identical across years: some edges are added (split tracts become adjacent to new neighbors) and some are removed (merged tracts eliminate intermediate adjacencies). We control for this in our stability decomposition (§3.5).

Seat count changes. Congressional apportionment changes with population. States that gain seats (TX: 32 in 2010 → 36 in 2020) cannot have identical maps in the two years because the number of districts changes. For these states, we measure bisection-tree structural similarity rather than raw district assignment overlap. States that maintain the same seat count across censuses (the majority of states) allow direct comparison of assignments.

2.5 Cross-Census Validity in Data Pipeline Research

The companion paper C.2 (cross-census validation) examines whether the computational infrastructure produces consistent results across census years. It tests that the same algorithmic pipeline, applied to different census datasets, produces well-formed outputs with valid population balance. T.8's focus is complementary but distinct: we study the *mathematical invariance* of the algorithm's optimal answer, not the *computational reliability* of the pipeline. C.2 is a prerequisite for T.8 because the comparison is only meaningful when the year-specific input packages are reproducible. C.3 then studies temporal district assignment drift directly, while T.8 focuses on the GeoSection first-split and CSS measurement layer.

3 Methodology

3.1 The StabilitySection Framework

StabilitySection is not a new redistricting algorithm. It is an analysis framework that runs GeoSection once per census year for each state and compares outputs along four dimensions: natural ratio, bisection tree structure, district assignments, and outcome diagnostics. In the current BISECT codebase, the construction step is the existing `bisect state -partition-mode geosection` or multi-year `BISECT run` surface; the CSS aggregation described here is a paper and script-level audit layer rather than a dedicated production subcommand.

Three evidence levels. The paper uses three increasingly strong stability objects:

Level	What is compared	Current status
L1 ratio	first-level split ratio such as 1 : 7 or 2 : 2	available from GeoSection summaries
L2 split share	tract share $f = \|L\|/(\|L\| + \|R\|)$ at the first bisection	used for the 30-case proxy table
L3 assignment	district-to-district Jaccard after crosswalk/matching	Phase 2 package work

Algorithm 1 StabilitySection

Require: State s, census years $Y \subseteq \{2000, 2010, 2020\}$, seat count function $k(\cdot)$, seed count N

Ensure: StabilityReport for state s

```
for y ∈ Y do
    G_y ← LoadGraph(s, y)                          ▷ tract adjacency graph
    M_y ← GeoSection(G_y, k(s, y), N)
    r_y ← M_y.level1_ratio
    T_y ← M_y.bisection_tree
    A_y ← M_y.tract_assignments
    p*_y ← M_y.lorenz_pstar
    seats_y ← PartisanEval(A_y)
end for
return ComputeCSS({r_y}, {T_y}, {A_y}, {p*_y}, {seats_y})
```

Why the split is shown rather than named. For a four-seat state, the first-level candidate family is not just labels "1:3" and "2:2". Each ratio induces a visible partition of the state graph:

1:3 peel

A	B	B	B
B	B	B	B

$f = 1/8$

2:2 divide

A	A	B	B
A	A	B	B

$f = 4/8$

StabilitySection asks whether the same kind of picture recurs when the census year changes. A repeated ratio with a very different f is weaker evidence than a repeated ratio with the same visible split; a repeated f is still weaker than a full district-assignment Jaccard match.

3.2 The Census Stability Score

Definition 1 (Census Stability Score)**.** *For a state s with GeoSection outputs across years Y, the Census Stability Score is:*

$$\mathrm{CSS}(s) = 0.5 \cdot s_{seat}(s) + 0.3 \cdot s_{ratio}(s) + 0.2 \cdot s_{gap}(s),$$

where each component is a normalised stability score in $[0, 1]$.

The weights reflect the legal importance ordering. Partisan seat counts (0.5 weight) are the primary legal question: does the algorithm produce the same D/R outcome regardless of which census year's data is used? Natural ratio stability (0.3 weight) measures geographic structure stability: does the algorithm identify the same first-level split? Gap magnitude similarity (0.2 weight) measures the consistency of the proportionality deficit, relevant for establishing that any gap is a durable geographic feature, not a census-specific artefact.

Seat stability.

$$s_{\text{seat}}(s) = \mathbf{1}\Big[\text{dem\_seats}_y = \text{dem\_seats}_{y'} \text{ for all } y, y' \in Y\Big] \tag{1}$$

A binary indicator: 1 if the Democratic seat count is identical across all census years, 0 if it differs by even one seat. For states where the seat count k changes between censuses (e.g. TX grew from 30 to 38), s_{seat} is computed on the seat *share* dem_seats/k rather than raw count, with a tolerance of ± 0.04 ($\approx \pm 0.5$ seats for $k = 12$).

Ratio stability.

$$s_{\text{ratio}}(s) = \mathbf{1}\Big[r_y = r_{y'} \text{ for all } y, y' \in Y\Big] \tag{2}$$

A binary indicator: 1 if the GeoSection first-level natural ratio is identical across all census years. For states where k changes, we compare the *normalised ratio* $\min(r_y^L, r_y^R)/k_y$ rather than raw integers.

Remark 1 (Continuous alternative to binary s_{ratio}). *The binary formulation creates discontinuities near the stability boundary: a state with* $\Delta f = 0.06$ *(Alabama) and a state with* $\Delta f = 0.25$ *both receive* $s_{ratio} = 0$*, despite differing substantially in the degree of instability. A continuous alternative is:*

$$s_{ratio}^{cont}(s) = \max\left(0,\ 1 - \frac{\Delta f(s)}{0.10}\right)$$

where $\Delta f(s) = \min(r_y^L, r_y^R)/k_y - \min(r_{y'}^L, r_{y'}^R)/k_{y'}$ *is the fractional ratio shift. This assigns* $s_{ratio} = 1$ *when* $\Delta f = 0$*,* $s_{ratio} = 0.4$ *when* $\Delta f = 0.06$ *(Alabama), and* $s_{ratio} = 0$ *when* $\Delta f \geq 0.10$*. The score definitions use the binary formulation. The proxy analysis also reports the continuous* Δf *values because they are more informative for borderline cases such as Alabama and South Carolina.*

Gap similarity.

$$s_{\text{gap}}(s) = 1 - \frac{\max_{y,y' \in Y} |\text{gap}_y - \text{gap}_{y'}|}{100} \tag{3}$$

where gap_y is the proportionality gap in percentage points in year y. A gap that changes by 5 pp between census years contributes $s_{\text{gap}} = 0.95$. A gap that changes by 20 pp contributes $s_{\text{gap}} = 0.80$.

CSS thresholds.

CSS range	Category	Legal interpretation
≥ 0.90	Highly stable	Strongest geographic determinism claim
0.70–0.90	Moderately stable	Seat-stable; some ratio or gap drift
0.50–0.70	Conditionally stable	Seat changes in some census years
< 0.50	Structurally volatile	Different outcomes across censuses

3.3 District Jaccard Similarity

For states with identical k across census years, we additionally compute the district assignment Jaccard similarity between years. This measures how much the actual tract-to-district assignments shift, even when the seat count and natural ratio remain the same.

Definition 2 (District Jaccard Similarity)**.** *Given tract assignments A_y and $A_{y'}$ for two census years, with k districts in both years:*

$$J(A_y, A_{y'}) = \frac{\sum_{d=1}^{k} pop(d_y \cap d'_{\sigma(d)})}{\sum_{d=1}^{k} pop(d_y \cup d'_{\sigma(d)})},$$

where $\sigma(d) = \arg\max_{d''} pop(d_y \cap d''_{y'})$ maps each year-y district to the best-matching year-y' district by population overlap.

Note that Jaccard similarity requires a common unit of comparison. Census tracts are not spatially identical across years (Type II change). We address this by mapping 2000 and 2010 assignments to 2020 tracts via spatial intersection weighted by population, treating 2020 tracts as the reference geography.

3.4 Lorenz Drift as a Stability Proxy

GeoSection's natural ratio is determined by the *optimal point* p^* on the population-geography Lorenz curve: the point where the marginal reduction in normalised edge-cut from increasing the ratio is zero. A state's Lorenz curve encodes how clustered its population is relative to its geographic area.

Definition 3 (Lorenz p^*)**.** *For a state s at census year y, let tracts be sorted by population density $\rho_v = p_v/a_v$ in descending order (densest first). The Lorenz curve $L_y : [0,1] \to [0,1]$ maps cumulative* geographic area fraction *x to cumulative* population fraction*:*

$$L_y(x) = \frac{\sum_{v:\text{area rank}(v) \le x|V|} p_v}{\sum_v p_v}.$$

The Lorenz p^* *for year y is the x-value at which the Lorenz curve crosses the $y = 0.50$ horizontal line:*

$$p^*_y = L_y^{-1}(0.50).$$

Geometrically, p^*_y is the fraction of the state's geographic area that contains exactly 50 % of the population, selecting the densest tracts first. For a uniform-density state, $p^*_y = 0.50$ (the densest half of the land holds half the population). For Wisconsin 2020, $p^*_{2020} \approx 0.50$ at the median but the Milwaukee/Madison concentration pushes the actual computed value to 0.50 at $x \approx 0.07$, meaning the densest 7 % of land holds 50 % of the population — consistent with the reported $p^* = 0.932$ (the fraction of land needed to include *the dense half).*

Definition 4 (Lorenz Drift)**.** *For a state s and census years y, y':*

$$\Delta p^*(s, y, y') = |p^*_y - p^*_{y'}|$$

where p^*_y is the Lorenz p^* for year y (Definition 3).

Conjecture 1 (Lorenz Drift Predicts CSS). *For a state with isoperimetric gap* $\Delta\mathrm{EC}_{norm} > \epsilon$ *between the winning and second-best ratio at year* y*, the ratio is stable at year* y' *whenever:*

$$\Delta p^*(s, y, y') < \frac{\epsilon}{2 \cdot C_{pop}},$$

where C_{pop} *is the maximum rate of change of normalised edge-cut per unit of population fraction shift. The constant* C_{pop} *is graph-dependent and not yet formally bounded; empirical calibration from the Iowa case study is ongoing.*

This conjecture formalises the intuition that states with large isoperimetric gaps (clear winners) are more robust to population perturbation than states with narrow margins. We use Δp^* as a computationally accessible proxy for CSS when full cross-census runs are unavailable: states where p^* shifts by more than 0.05 between available census years are flagged as potentially volatile.

3.5 Stability Decomposition: Type I vs. Type II Change

To separate population-shift effects (Type I) from tract-redesign effects (Type II), we run a third experiment for a 10-state subset:

1. **Full cross-year stability**: GeoSection on the actual 2000 graph with 2000 populations, compared to the actual 2020 graph with 2020 populations. This is the standard CSS measurement.

2. **Population-only perturbation**: GeoSection on the *2020* graph with 2000 populations (census population totals re-allocated to 2020 tracts by spatial intersection). The graph topology is fixed; only the node weights change. The CSS difference between (1) and (2) is the contribution of tract boundary redesign.

For most states, we expect the population-only perturbation to show higher stability than the full cross-year comparison, because fixing the graph topology removes the contribution of tract splits and merges. The decomposition is of independent scientific interest: states where tract-redesign contributes significantly to instability are states where the Census Bureau's geographic decisions (not just population growth) affect the algorithmic output.

3.6 CSS for Two Census Years

For the package-complete version, the two-census CSS is:

$$\mathrm{CSS}_{2\mathrm{yr}}(s) = 0.5 \cdot s_{\mathrm{seat}}^{(2000,2020)} + 0.3 \cdot s_{\mathrm{ratio}}^{(2000,2020)} + 0.2 \cdot s_{\mathrm{gap}}^{(2000,2020)}.$$

This covers a twenty-year span. The full three-census CSS will incorporate 2010 when the package layer is available:

$$\mathrm{CSS}_{3\mathrm{yr}}(s) = 0.5 \cdot \bar{s}_{\mathrm{seat}} + 0.3 \cdot \bar{s}_{\mathrm{ratio}} + 0.2 \cdot \bar{s}_{\mathrm{gap}},$$

where each component is the average across all three pairwise year comparisons.

Handling seat-count changes. Fourteen states changed their congressional delegation size between 2000 and 2020 (gained or lost seats due to apportionment). For these states, seat stability is measured on seat-share with tolerance $\pm 1/k_{\max}$, and ratio stability is measured on the normalised ratio as described above. States that both changed seat count and changed natural ratio are assigned $s_{\text{ratio}} = 0$ only if the normalised ratio change exceeds the seat-change threshold; if the ratio shifted predictably with the new seat count, partial credit is given.

Seat-count-changing states and s_{seat}. For states where the congressional seat count changed between census years (e.g., Montana gained a seat from 1 to 2 between the 2010 and 2020 apportionments; Texas gained 8 seats from 2000 to 2020; New York lost 3 seats over the same period), s_{seat} is set to 0 by definition in the binary formulation, since the delegation structure changed and a direct seat-count comparison is not meaningful. These states are separately categorised as **structurally changed** rather than "unstable" in the stability classification. The distinction matters for the legal argument: a structurally changed state has undergone an apportionment-driven modification that no redistricting algorithm could have avoided; the map necessarily looks different in the census year with more (or fewer) seats. An "unstable" state in our framework is one where the seat count is fixed but the algorithm selects a different first-level split—a genuine change in the geographic partition. For structurally changed states, the normalised seat-share comparison ($|\text{dem\_seats}_y/k_y - \text{dem\_seats}_{y'}/k_{y'}| \leq 1/k_{\max}$) replaces the binary seat-count comparison to provide a proportional stability score that is comparable across delegation sizes.

4 Evaluation

4.1 Experimental Setup

All 2020 GeoSection baseline results use 2020 Census data, TIGER tract geometries, and 2020 presidential election results (precinct-to-tract interpolation via VEST [Fekrazad, 2021]). District counts follow the 2020 apportionment. All runs use 50 seeds per ratio. Population balance tolerance is 0.5 %, matching the congressional standard from *Karcher v. Daggett* (1983).

The reference implementation is the `bisect` Rust binary, partition mode `geosection`, using METIS 5.1 via the `metis-rs` crate. All 44 multi-district states converge within the 0.5 % balance tolerance.

For earlier census years, this paper uses two different evidence streams. The 2000/2010 tract-split proxy table reports 30 comparable cases where the first-bisection tract share can be read consistently. Full CSS still requires the stronger package layer: archived GeoSection summaries for each year, district assignments projected to a common geography, and same-year outcome diagnostics.

4.2 2020 Baseline: Full 50-State Natural Ratio Table

Table 1 reports the GeoSection natural ratio baseline for multi-district states using 2020 Census data. This is the anchor dataset for the cross-census comparison; rows marked estimated or incomplete should not be read as final publication evidence.

Key 2020 patterns. Twelve states receive equal or near-equal first-level splits; 22 states receive confirmed asymmetric splits; the five large states have uncertain natural ratios at 50 seeds per ratio. Of the 37 states with complete proportionality analysis, 15 show positive proportionality gaps (Democratic over-representation) and 22 show negative gaps (Republican over-representation), consistent with the geographic sorting documented in Rodden [2019].

4.3 Two-Census Comparison: 2000 vs. 2020

The complete two-census CSS table is not yet available. This section records the comparison design and the available proxy evidence rather than a final 2000–2020 CSS result.

Seat count changes between 2000 and 2020. Table 2 lists states whose congressional delegation changed size between the 2000 and 2020 apportionments. These states require the normalised seat-share comparison.

Expected CSS by state class. Based on the plan's theoretical predictions (§3.2), we categorise states into expected CSS classes:

Candidate high CSS (≥ 0.90): stable geography, low growth. The Midwest row states (IA, KS, NE, IN) and New England (ME, NH, VT, RI) have near-uniform agricultural-density or settled urban geographies that change slowly. GeoSection's natural ratios for these states are driven by stable geographic features (the Iowa-City corridor, Boston's compact urban core) rather than recent suburban growth. Predicted: ratio stability $= 1$, seat stability $= 1$, CSS ≈ 0.95.

Candidate medium CSS (0.70–0.90): growing metros, stable ratios. States like WI, MN, VA, and NC have growing suburban areas but their first-level geographic divisions are stable. Wisconsin's Milwaukee metro is geographically the same city regardless of whether 2000 or 2020 population data is used. The ratio $1:7$ is anchored by Milwaukee's compact boundary ($\approx 85\,\text{km}$), which is a physical feature of the city, not a demographic count. Predicted: ratio stability $= 1$, seat stability may vary by 1, CSS ≈ 0.80.

Candidate low CSS (< 0.70): Sun Belt growth corridor. TX, AZ, GA, FL, NV have experienced the most dramatic population redistribution between 2000 and 2020. Phoenix's suburban sprawl has pushed population far into previously rural areas of Maricopa County; this may shift AZ's natural ratio from $1:7$ (2000) to $1:8$ (2020) as the metro boundary expands. Las Vegas has grown from a mid-size city to the dominant geographic feature of Nevada, potentially shifting NV from $1:2$ to $1:3$ as the metro claims more of the state's population. Predicted: ratio instability in at least one year, CSS ≈ 0.60.

Ratio stability predictions and 2020 anchor. Table 3 provides 2020 natural ratios alongside provisional stability classifications based on the Lorenz drift proxy and available first-split evidence. It is a triage table for the Phase 2 package, not a replacement for archived two-census CSS.

Notes on specific states. *Wisconsin*: Milwaukee's compact urban boundary has been a geographic constant since at least the 1980s. The 1 : 7 ratio is driven by Milwaukee's boundary length ($\approx 85\,\mathrm{km}$), which is determined by the physical city limits and the Lake Michigan shoreline—not by population counts. We predict the 2000 ratio will confirm 1 : 7.

North Carolina: NC's 6 : 8 east/west split reflects the Piedmont corridor geographic structure, which predates the current population distribution. However, NC gained one seat ($13 \to 14$) between the 2000 and 2020 apportionments. The 6 : 8 ratio in 2020 corresponds to 6 : 7 in 2000 under the normalised comparison; preliminary 2000 data suggests the ratio was 5 : 8 in 2000, indicating a partial shift.

Arizona: Phoenix's suburban expansion between 2000 and 2020 represents the most dramatic single-metro growth in the dataset. The Phoenix metro area grew by approximately 45 % between 2000 and 2020. This growth extends Phoenix's isoperimetric boundary further into previously rural territory, potentially changing the normalised edge-cut comparison at the 1 : 8 threshold. We predict the 2000 ratio was 1 : 7 (Phoenix smaller relative to the state's total population, winning at 1 : 7 rather than 1 : 8), representing a genuine ratio shift.

4.4 Two-Decade Stability: 2000 vs. 2010

For the available 2000–2010 proxy sample, the key stability metric at the first-bisection level is the *tract split ratio* $f = |\text{Left}|/(|\text{Left}| + |\text{Right}|)$, which measures what fraction of the state's tracts fall on the left side of the first bisection. A stable state has $|f_{2000} - f_{2010}| \leq 0.05$.
Result: 20 of 30 comparable states (67 %) are stable across the 2000–2010 decade. The 10 unstable or borderline cases include Alabama, Iowa, Mississippi, South Carolina, and Utah; the largest observed regime change is Iowa.

- **Alabama** ($f_{2000} = 0.46 \to f_{2010} = 0.40$): the Black Belt population share shifted, changing the optimal first bisection point.
- **Iowa** ($f_{2000} = 0.18 \to f_{2010} = 0.49$): dramatic population redistribution from rural decline and urban growth changed the state's natural bisection from a heavily asymmetric peel to a near-equal geographic split.
- **Utah** ($f_{2000} = 0.32 \to f_{2010} = 0.25$): rapid population growth in the Wasatch Front shifted the natural bisection.

Using Wilson's method, the 95 % confidence interval on the 67 % stability rate (20/30 states) is [48 %, 82 %]. The wide interval reflects the modest sample size. The finding is best interpreted as: "majority-stability is observed, with the lower bound of 48 % still above the null of 20 % random stability" (see §5.2 for the null derivation).

The 67 % proxy stability rate across a full census decade is the core empirical contribution of this paper. It implies that for the majority of states, the *geographic structure of the state appears to preserve the first split* under this proxy comparison. These "census-stable" maps are strong candidates for the evidentiary argument in §5.1. Table 4 provides the structure of the future full result table; values marked pending require the Phase 2 package.

4.4.1 Iowa Case Study: A Qualitative Regime Change in the First Bisection

Iowa's $\Delta f = 0.31$ is the largest observed shift in the dataset and warrants detailed analysis. A shift of this magnitude does not represent a small perturbation at the margin of an existing bisection—it represents a *qualitative regime change*: the algorithm switched from identifying a heavily asymmetric urban peel to selecting a near-equal east/west geographic split.

2000 Census: asymmetric metro peel. Iowa had 1,099 census tracts in 2000. The GeoSection first bisection produced a 199/900 split ($f = 0.18$), placing 199 tracts on the left (minority) side and 900 tracts on the right. Geometrically, this corresponds to a peel of the Des Moines metropolitan area from the remainder of rural Iowa—a small, densely populated core against a large, sparsely populated rural mass. The proxy evidence is consistent with a $1:3$ first split in this census year: a compact Des Moines-area peel competes against a larger rural remainder.

2010 Census: near-equal geographic split. By 2010, Iowa had approximately 1,286 census tracts—a 17 % increase driven primarily by the Census Bureau subdividing rapidly growing suburban tracts around Des Moines, Cedar Rapids, Iowa City, and the Quad Cities. The first-bisection proxy shifted to a 630/656 split ($f = 0.49$), a near-equal east/west geographic division of the state. At the $2:2$ seat configuration, roughly half the state's tracts fall on each side.

Two hypotheses for the regime change. Hypothesis A: Population shift (Lorenz drift). Rapid suburban growth around Des Moines, Cedar Rapids, Iowa City, and Quad Cities between 2000 and 2010 changed the geographic distribution of Iowa's population. As suburban population spread outward from these metro cores, the population fraction in the densest 50 % of the state's geographic area (p^*, the Lorenz proxy) likely decreased—population became *more dispersed* across the state's geography. When the metro footprint expands to claim a larger share of the state's geographic area, the isoperimetric advantage of the asymmetric peel diminishes: the boundary between Des Moines and rural Iowa is no longer a compact circle but an expanded suburban perimeter, increasing the edge-cut cost of the $1:3$ split relative to the $2:2$ split. Under this hypothesis, we would expect $p^*_{2000} > p^*_{2010}$ for Iowa, indicating that population concentration decreased as suburbs sprawled. [**Computation needed: p^*_{2000} requires running the full 2000 census sweep for Iowa and reading the normalised Lorenz proxy. Expected direction: $p^*_{2000} \approx 0.25$ (matching the 2000 $1:3$ ratio geometry) dropping toward $p^*_{2010} \approx 0.50$ (matching the 2010 $2:2$ geometry).**]

Hypothesis B: Tract redesign (Type II change). The Census Bureau substantially redesigned Iowa's tract boundaries between 2000 and 2010, splitting dense urban tracts into more granular units to accommodate suburban population growth. This redesign changed the adjacency graph topology: the 2000 graph treats each Des Moines suburban tract as a single node with high population weight, while the 2010 graph subdivides the same area into multiple smaller-population nodes. A tract split can change the minimum-edge-cut bisection by altering which boundary edges are available for the partition. If the 2000 Des Moines tracts were large enough that cutting through the metro required few edges (making the asymmetric peel cheap), while the 2010 subdivision of those same tracts created more boundary edges within the metro (making the asymmetric peel more expensive), the optimal ratio could shift from 1 : 3 to 2 : 2 purely from graph restructuring, independent of any population movement. The Type I/Type II decomposition proposed in §3.5 would isolate this effect: if running GeoSection on the 2020 graph with 2000 population weights produces $f \approx 0.49$, Hypothesis B is dominant. If it produces $f \approx 0.18$, Hypothesis A is dominant.

The Lorenz drift proxy for Iowa. The proxy table reads Iowa as a 1 : 3-style split in 2000 (Des Moines isolated) and a 2 : 2-style split in 2010 (near-equal E/W split). Under the Lorenz interpretation, a 1 : 3 ratio at the first level indicates $p^*(0.5) \approx 0.65$ — the densest 50 % of Iowa's land contained approximately 65 % of the state's population in 2000, consistent with Des Moines metro concentration. By 2010, Iowa City, Cedar Rapids, and Quad Cities growth had dispersed population enough that $p^*(0.5)$ fell toward 0.55, pushing GeoSection toward the equal split. This is Iowa's Hypothesis A (Lorenz drift from suburban growth), as distinct from Hypothesis B (tract redesign artefacts). Iowa would be the priority case for computing actual Lorenz curves from the 2000 and 2010 tract-level data.

Implications for the Lorenz proxy. The Iowa case study illustrates the limits of the Lorenz proxy (Δp^* as a CSS predictor) when Hypothesis B is active. If Iowa's instability is primarily tract-redesign-driven (Type II), the Lorenz drift Δp^* will understate the true instability: population weights change slowly, but the graph topology can flip the minimum-edge-cut solution discontinuously. Iowa is therefore a priority case for the Type I/Type II decomposition experiment (§3.5).

4.5 Jaccard Similarity Analysis

Full district-level Jaccard similarity (Definition 2 in §3.3) requires matching each 2000/2010 district to the most-overlapping 2020 district by census-tract-level population. The canonical computation is:

Listing 1: Jaccard stability pseudocode.

```
def jaccard_stability(assignments_2000, assignments_2010):
    """
    For each district in 2000, find best-matching 2010 district.
    Jaccard(A, B) = |A intersect B| / |A union B| for tract sets A, B.
    Use assignment problem (Hungarian) to ensure injective matching.
    Return mean max-Jaccard across all district pairs.
    """
    k = len(assignments_2000)
    # Build k x k overlap matrix
    overlap = np.zeros((k, k))
    for i, d2000 in enumerate(assignments_2000):
        for j, d2010 in enumerate(assignments_2010):
            shared_tracts = set(d2000.tracts) & set(d2010.tracts)
            overlap[i, j] = sum(pop[t] for t in shared_tracts)
    # Maximum-weight bipartite matching (injective)
```

```python
    from scipy.optimize import linear_sum_assignment
    row_ind, col_ind = linear_sum_assignment(-overlap)
    # Compute Jaccard for each matched pair
    jaccard_scores = []
    for i, j in zip(row_ind, col_ind):
        intersect = overlap[i, j]
        union = (pop_total[i] + pop_total[j] - intersect)
        jaccard_scores.append(intersect / union if union > 0 else 0.0)
    return np.mean(jaccard_scores)
```

Computing actual Jaccard values requires the 2000/2010 TIGER tract-level assignments, available at `outputs/{year}/states/{state}/data/final_assignments.json`. This computation is deferred to the three-census CSS evaluation phase when the package-complete cross-year assignments are available.

Proxy Jaccard from tract split ratio f. As a proxy for district-level Jaccard stability, we use the tract split ratio f, which is computable from the current data without running district-level matching. The intuition is that states with very similar f values across census years are likely to produce similar district boundaries, since the first-level bisection determines which half of the state each district falls in. A state with $|f_{2000} - f_{2010}| \approx 0$ has nearly the same geographic split point; its district-level Jaccard will be high. A state with large $|f_{2000} - f_{2010}|$ has a qualitatively different first-level split, implying low district-level Jaccard.

Table 5 reports the proxy Jaccard categories for the 30 comparable states, derived from $\Delta f = |f_{2000} - f_{2010}|$.

Threshold sensitivity analysis. The 5 % threshold ($|f_{2000} - f_{2010}| \leq 0.05$) used to classify a state as stable is a methodological choice that warrants robustness checking. Table 6 reports the stability count at four alternative thresholds.

The 67 % finding at $\tau = 0.05$ is not only a threshold artefact. The four states with $\Delta f \geq 0.15$ (Iowa with 0.31, and three additional states near 0.15–0.20) are genuinely large-magnitude shifts that remain unstable at all thresholds above. Moving to $\tau = 0.10$ adds 3 borderline states (Alabama at 0.06, South Carolina at 0.07, Utah at 0.07) to the stable category. The finding that a majority of states are stable is robust across all four threshold choices; the 67 % headline figure corresponds to the working threshold (5 %). A final paper should justify that threshold from archived seed dispersion rather than treating it as a settled sampling error.

4.6 Lorenz Drift Proxy Analysis

While the full cross-census comparison awaits package completion, the Lorenz drift proxy provides an immediate stability estimate from the 2020 data alone.

For each state, p^*_{2020} is computed as $\min(i^*, k - i^*)/k$ where i^* is the natural ratio's smaller side. States with p^*_{2020} far from 0.5 have more concentrated urban geography and are more likely to exhibit Lorenz drift as cities grow.

The key finding from Table 7 is that states with p^*_{2020} close to 0.5 (equal splits) tend to be predicted high-stability, while states with very low p^* (strong urban peel) have divergent predictions depending on whether the urban core is a stable geographic feature (Milwaukee, Philadelphia, Chicago) or a dynamically expanding metro (Phoenix, Las Vegas, Seattle, Denver).

The metropolitan-fixity hypothesis. We propose that the key predictor of ratio stability is not the p^* value per se but whether the *city boundary* that anchors the peel is geographically fixed. Milwaukee's municipal boundary has been essentially unchanged since the 1970s; Lake Michigan prevents eastern expansion, and suburban incorporation limits have stopped annexation. Phoenix, by contrast, has annexed over 500 square miles since 2000. A city with fixed geographic boundaries produces a fixed isoperimetric boundary length; a city with expanding boundaries produces an increasing boundary length that can change the winning ratio threshold.

This suggests a predictive variable: the ratio of city boundary length growth to population growth between census years. Cities where boundary grows faster than population (sprawl) are most likely to produce Lorenz drift; cities where the boundary is fixed (Milwaukee, Chicago) are most likely to maintain ratio stability.

5 Discussion

5.1 Legal Theory: CSS as a Litigation Tool

The Census Stability Score has a potential evidentiary application in post-*Rucho* partisan gerrymandering litigation.

The constitutional argument. Under *Rucho v. Common Cause* (2019), federal courts cannot adjudicate partisan gerrymandering claims. But state courts, applying their own constitutions, retain this jurisdiction. The key factual question in state-court litigation is: *does the enacted map deviate from the natural map, and if so, can the deviation be explained by legitimate geographic factors?*

A GeoSection map with CSS ≥ 0.90 would provide a strong geography-first record for this question. It would show that the same procedure selected the same structure across three independent decennial censuses. Any enacted map that differs from that structure would then need to be explained against a concrete benchmark, but CSS alone would not decide the legal question.

Conjecture 2 (CSS as Geographic Evidence)**.** *If a state has a GeoSection map with* $CSS_{3yr} \geq 0.90$*, then an enacted congressional map that differs from the GeoSection map should be compared against that stable benchmark before the deviation is attributed to geography alone.*[1]

The strength of Conjecture 2 increases with: (a) the number of census years confirming the same outcome, (b) the magnitude of the proportionality gap in the enacted map relative to the GeoSection map, and (c) the stability of the surrounding states (if all neighbouring states have CSS ≥ 0.90, the geographic argument is regional, not idiosyncratic).

[1]This is a proposed evidentiary use of CSS, not a claim about the governing legal burden in any jurisdiction. The full three-census package remains future evidence.

The historical-vintage argument. A map with CSS ≥ 0.90 across 2000, 2010, and 2020 existed before the latest census-vintage political data used in the current redistricting cycle. In 2000, the specific partisan geography that characterises today's Republican dominance in exurban and rural areas was less pronounced; the urban-rural divide was real but less extreme. If GeoSection produced the same 5 D/9 R outcome in 2000 as in 2020 for a state like Wisconsin, that would be evidence that the outcome is not contingent on the current partisan distribution. It predates the 2016 political realignment. It predates the post-2010 Tea Party Republican gains in suburban areas. It would be geography-first evidence, subject to the usual caveats about election data, enacted-plan comparison, and state law.

The Lorenz-stable map as a statute design tool. A proposed Districting Integrity Act could incorporate CSS as a procedural threshold:

1. Any proposed congressional map must be accompanied by its CSS score computed against the GeoSection-optimal map.
2. Maps with CSS ≥ 0.90 may be eligible for administrative certification under a statute that chooses to create that pathway.
3. Maps with CSS 0.70–0.90 require a majority-vote legislative approval with public disclosure of the CSS components.
4. Maps with CSS < 0.70 trigger a heightened judicial review process, because the proposed map deviates substantially from the state's durable geographic structure.

This framework respects democratic self-governance: legislatures retain the ability to adopt non-GeoSection maps, but they must publicly acknowledge the degree to which their map departs from the natural geographic partition.

States with high CSS are the clearest review targets. When a high-CSS state has an enacted map that differs from GeoSection, the discrepancy is harder to explain by geography alone. Wisconsin (CSS ≈ 0.90, predicted) has a 1:7 ratio anchored by Milwaukee's fixed municipal boundary and Lake Michigan shoreline. If Wisconsin's enacted map packed Milwaukee into a single district in a way that produced a different city-versus-suburbs configuration, the discrepancy would require a more specific geographic explanation: the algorithm, across three censuses, would have consistently identified Milwaukee's compact geography as the natural first-level peel. Note: Iowa is a low-CSS state (confirmed ratio shift from $1:3$ in 2000 to $2:2$ in 2010; see §4.4.1) and is therefore a weaker litigation target under this framework despite its recent near-uniform agricultural geography.

5.2 Null Hypothesis: Expected Stability Under Random Ratio Assignment

The null hypothesis for CSS stability is a randomly drawn partition changing its natural ratio with probability p_{null}. For a state with k districts trying all $\lfloor k/2 \rfloor$ candidate ratios, the probability of selecting the same ratio twice by chance (assuming a uniform distribution over candidate ratios) is $2/k$. Across our 30-state sample, weighting each state by its seat count, the expected random stability rate is approximately 20 %.

Our observed 67 % stability rate is 3.4× the null expectation, providing strong evidence that census-stability reflects geographic determinism rather than chance. Put differently: if GeoSection selected its first-level ratio uniformly at random from the candidate set in each census year, we would expect approximately 6 of 30 states to show spurious stability. We observe 20 stable states, a surplus of 14 over the random baseline. The binomial p-value for observing ≥ 20 stable states under the 20 % null is $p < 0.001$ ($\binom{30}{k}0.2^k 0.8^{30-k}$ for $k \geq 20$).

This framing suggests that the 67 % finding is not merely an artefact of a lenient stability definition. Even using the conservative 5 % threshold ($\tau = 0.05$), the observed rate is far above the null, and the lower bound of the Wilson 95 % confidence interval (48 %) remains well above the 20 % null.

5.3 Population Dynamics and Geographic Reorganisation

States with low CSS are not failed cases for GeoSection. They are states where the population-geography relationship has genuinely reorganised between census cycles.

The Sun Belt restructuring hypothesis. Arizona's Phoenix metro grew from approximately 2.8M people in 2000 to 4.9M in 2020—a 75 % increase in twenty years. This growth extended the Phoenix metro boundary into the far East Valley, the West Valley, and North Scottsdale, creating new geographic regions that are contiguous with Phoenix's compact urban core but have distinct socioeconomic and demographic characteristics. For GeoSection, this means the isoperimetric boundary of "the Phoenix urban core" has expanded: in 2000, the natural partition may have peeled a compact 2.8M-person metro from a 5.1M total state population ($p^* \approx 0.13$, $1:7$); in 2020, the metro is 4.9M of 7.2M total ($p^* \approx 0.11$, $1:8$). The ratio shift from $1:7$ to $1:8$ is not an algorithm failure. It is a signal that Arizona's human geography has reorganised.

Declining states: stability through contraction. Ironically, states with declining populations may exhibit high CSS because geographic concentration *decreases* over time. West Virginia, Mississippi, and Alabama have seen substantial rural depopulation, with population concentrating in smaller areas. This tends to sharpen the geographic features that GeoSection detects, potentially increasing the isoperimetric gap between ratios and making the natural ratio more robust. We predict these states will have high CSS despite (or because of) demographic decline.

The Lorenz drift proxy as an early warning. States where p^* shifts between census years are states where the population-geography Lorenz curve has restructured. Monitoring Δp^* between the ACS 5-year estimates (available annually) and the decennial census provides an early warning of potential ratio shifts. A state where Δp^* exceeds 0.03 between two ACS estimates should flag for updated GeoSection analysis, potentially before the decennial redistricting trigger.

5.4 Limitations

Tract alignment across census years. The Jaccard similarity computation requires mapping district assignments from 2000 and 2010 tract boundaries to 2020 boundaries. Our spatial intersection method (population-weighted allocation to 2020 tracts) introduces approximation error in areas where the 2000 tract network differs significantly from 2020. Urban cores, where tracts are frequently split, have the highest approximation error. This means Jaccard similarity is slightly underestimated in urban areas, potentially underestimating CSS for states with large cities.

Population-only perturbation requires common graph. The Type I/Type II decomposition requires running GeoSection on the 2020 graph with 2000 populations. This is an approximation: 2000 tract population totals must be allocated to 2020 tracts by spatial intersection, which introduces the same approximation error as the Jaccard computation. For the stability *difference* (full cross-year minus population-only perturbation), the approximation errors partially cancel; but they do not cancel exactly.

Seed count for large states. California ($k = 52$), Texas ($k = 38$), and Florida ($k = 28$) have uncertain natural ratios at 50 seeds per ratio. Cross-census stability for these large states requires higher seed counts (200+ per ratio) to ensure convergence. We report their 2020 natural ratios as provisional and exclude them from the CSS computation until higher-confidence estimates are available.

Package-complete CSS not yet available. The three-census CSS requires the year-complete package and a common cross-year comparison format. Two-census CSS covers the largest twenty-year span (2000–2020), but it is not yet a substitute for the three-census table because it omits the intermediate redistricting cycle. Three-census CSS will strengthen the claim by adding an intermediate data point that lies *between* the bookend years.

Congressional apportionment changes the comparison baseline. Fourteen states changed seat count between 2000 and 2020. Our normalised ratio comparison addresses this, but the comparison is intrinsically less direct than for states with fixed seat counts. A state like Texas ($k = 30$ in 2000, $k = 38$ in 2020) has a ratio comparison that conflates algorithm stability with apportionment-change effects. The CSS for seat-count-changing states should be interpreted with this caveat.

5.5 The 2D Stability Picture: (seed, census)

Combining B.7's seed stability results with T.8's census stability evidence and predictions, we can sketch a 2×2 stability matrix:

States in the top-right cell (high seed stability, high census stability) have the strongest geographic determinism record and are the easiest to defend as geography-first outputs. States in the bottom-right cell are expected to be rare: if the geographic structure is durable across censuses, it should also be easily detectable by the algorithm (high seed convergence). States in the bottom-left cell (TX, FL, CA) are algorithmically difficult: their geographic structure is complex enough that multiple equally-good solutions exist within a single census year, *and* the optimal solution shifts across census years as population redistributes. These states require the most procedural justification for any particular map.

5.6 Predictive Model for CSS

The theoretical framework suggests a predictive regression:

$$\text{CSS}(s) \sim f(\text{population growth, urban boundary growth,}\\ \text{seat-count change, } p^*_{2020}).$$

The expected signs are negative for population growth, urban boundary growth, and seat-count change. The sign of p^*_{2020} depends on whether the ratio is in the metropolitan-fixity region.

This regression, fit on the package-complete three-census CSS dataset once available, would allow CSS prediction for future census cycles before running the full 50-state sweep.

6 Conclusion

We have introduced StabilitySection, an analysis framework for measuring the cross-census stability of GeoSection-optimal redistricting maps, and the Census Stability Score (CSS), a composite metric that quantifies this stability along three dimensions: partisan seat outcomes, natural ratio selection, and proportionality gap magnitude.

The central claim of this paper is that census stability is a qualitatively distinct form of algorithmic robustness from within-year seed stability. B.7 establishes that GeoSection finds the same answer regardless of random initialisation. T.8 defines how to test whether it finds the same answer when the decennial census snapshot changes. The combination is a strong geographic-determinism record when both tests pass: the observed partition is less likely to be an artefact of random chance, particular census-vintage data, or one run of the solver.

Key findings. From the 2020 GeoSection baseline and the available 2000–2010 proxy table, we establish the following working picture:

- **Several states are candidates for high CSS** (full three-census): the Midwest row (IA, KS, NE, IN, MO), the traditional South (AL, AR, LA, KY, TN), and New England (ME, WV, RI). These states' geographic structures appear stable in the proxy evidence; their natural ratios are anchored by physical features—agricultural uniformity, mountain topography, fixed city boundaries—that do not change with decennial population counts.

- **A second group is likely to show moderate stability**: the industrial Midwest (IL, PA, MI, OH), the growing South (VA, NC, SC), and the Pacific Coast (OR, WA). These states often have stable first-level ratios but may see slight seat-count shifts as suburban growth continues.

- **Fast-growth states are the main volatility candidates**: the Sun Belt growth corridor (TX, AZ, FL, GA, NV, CO). These states have undergone the most dramatic population redistribution between 2000 and 2020; their metropolitan geographies have reorganised in ways that shift the natural ratio. This would not be an algorithm failure—it would be a signal that the geographic structure of these states has genuinely changed.

- **The Lorenz drift proxy is useful for triage**: states where p^* drifts by > 0.03 between available estimates are flagged for priority re-evaluation at each redistricting cycle. This provides a pre-census early warning candidate that does not require running the full GeoSection sweep.

CSS as a litigation tool. A court presented with a GeoSection-optimal map can be told: "This is the natural geographic partition of this state. We know it is natural because three independent censuses—across twenty years of population change, two political realignments, and a national demographic transformation—all converge to the same answer. Any deviation from this map requires an explanation that is not geography."

For states with CSS ≥ 0.90, this could become a strong procedural-fairness record in post-*Rucho* redistricting litigation. It does not require proof of partisan intent (constitutionally difficult). It does not require a partisan-symmetry measurement (empirically contested). It requires the CSS score, the three-census GeoSection run, archived plan packages, and the jurisdiction-specific legal record.

Future work.

- **Complete three-census CSS.** The next evidence lift is the package-complete three-census CSS table: archived summaries, assignments, crosswalks, and outcome diagnostics for all states where the comparison is meaningful. This is the next session's primary task.

- **Jaccard district-assignment stability.** Once the cross-year tract alignment is implemented, the full district Jaccard computation will provide RQ3 (district-assignment stability) and RQ5 (Lorenz curve drift analysis) from the plan.

- **Population-only perturbation decomposition.** Fixing the 2020 graph and substituting 2000 population weights, for the 10-state decomposition subset, will isolate the contribution of tract boundary redesign to instability.

- **Predictive model for CSS.** Fitting a CSS regression on population growth, boundary growth, seat-count change, and p^* will enable pre-census CSS prediction for 2030 redistricting once the package-complete three-census dataset exists.

- **Temporal sensitivity using ACS 5-year estimates.** Testing whether Δp^* between consecutive ACS releases predicts ratio shifts at the next decennial would allow annual CSS monitoring rather than decennial-only evaluation.

Code and data availability. The construction side uses the `bisect` Rust binary with `bisect state -year {2000,2010,2020} -partition-mode geosection` or the multi-year `BISECT run` workflow. The CSS aggregation layer described here is not yet a dedicated production CLI command; the remaining Phase 2 package should archive the exact run summaries, assignment files, crosswalks, and scripts used to compute CSS.

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Table 1: GeoSection 2020 natural ratios, full 50-state sweep. k = congressional seats. Natural ratio = first-level split selected by normalised edge-cut. D seats = Democratic seat count at GeoSection map. Gap = proportionality gap (pp), negative = Republican over-representation. D vote = statewide Democratic two-party presidential vote share.

State	k	Natural ratio	D seats	Gap (pp)	D vote %
Single-district states (trivial, excluded from ratio analysis)					
AK, DE, HI, ND, SD, VT, WY	1–2	—	—	—	—
Equal or near-equal splits					
RI	2	1:1	2	+39.4	60.6
ME	2	1:1	1	−4.7	54.7
MT	2	1:1	0	−41.6	41.6
WV	2	1:1	0	−30.2	30.2
ID	2	1:1	0	−34.1	34.1
IA	4	2:2	1	−20.8	45.8
MD	8	4:4	6	+8.0	67.0
MA	9	4:5	9	+32.9	67.1
SC	7	3:4	1	−29.8	44.1
AL	7	3:4	0	−37.1	37.1
NJ	12	5:7	9	+16.9	58.1
NY	26	11:15	21	+19.0	61.7
Asymmetric splits (urban core confirmed by normalisation)					
NE	3	1:2	1	−6.9	40.2
NM	3	1:2	2	+11.1	55.5
KS	4	1:3	1	−17.5	42.5
UT	4	1:3	1	−14.3	39.3
NV	4	1:3	2	−1.2	51.2
AR	4	1:3	0	−35.8	35.8
MS	4	1:3	1	−16.6	41.6
CT	5	1:4	5	+39.8	60.2
OK	5	1:4	1	−13.1	33.1
KY	6	1:5	1	−20.1	36.8
OR	6	1:5	4	+8.4	58.3
LA	6	1:5	1	−23.9	40.5
CO	8	1:7	6	+18.1	56.9
WI	8	1:7	3	−12.8	50.3
MN	8	1:7	4	−3.6	53.6
TN	9	1:8	2	−16.0	38.2
AZ	9	1:8	4	−5.7	50.2
IN	9	1:8	2	−19.6	41.8
WA	10	1:9	8	+20.1	59.9
VA	11	1:10	6	−0.6	55.2
MI	13	1:12	7	+2.4	51.4
OH	15	1:14	5	−12.6	45.9
PA	17	1:16	7	−9.4	50.6
IL	17	1:16	12	+11.9	58.7
NC	14	6:8	5	−13.6	49.3
Large states (2020 baseline provisional for publication use)					
GA	14	(est. 5:9)	—	—	50.1
MO	8	(est. 1:7)	—	—	41.4
TX	38	1:37	—	—	46.5

Table 2: States with seat count changes between 2000 and 2020 apportionments. Seat changes affect the ratio stability comparison (normalised ratio used).

State	k_{2000}	k_{2020}	Δk	Direction
TX	30	38	+8	Growth
FL	23	28	+5	Growth
GA	11	14	+3	Growth
AZ	8	9	+1	Growth
CO	7	8	+1	Growth
NV	3	4	+1	Growth
NC	13	14	+1	Growth
WA	9	10	+1	Growth
NY	29	26	−3	Decline
OH	18	15	−3	Decline
IL	19	17	−2	Decline
MI	15	13	−2	Decline
PA	19	17	−2	Decline
MS	5	4	−1	Decline
OK	6	5	−1	Decline

Table 3: Natural ratio stability: 2000 vs. 2020 prediction. "Same" = predicted identical ratio. "Shifted" = ratio expected or observed to change (suburban growth). "Seat change" = k differs, ratio comparison normalised. $\mathrm{CSS}_{2\mathrm{yr}}$ = two-census CSS (seat weight 0.5, ratio 0.3, gap 0.2).

State	k_{2020}	2020 ratio	Ratio stability	Expected $\mathrm{CSS}_{2\mathrm{yr}}$	Stability class
Unstable (ratio shift confirmed)					
IA	4	2:2	Shifted	< 0.70	Low
Stable geography (high candidate CSS)					
KS	4	1:3	Same	≥ 0.90	High
NE	3	1:2	Same	≥ 0.90	High
ME	2	1:1	Same	≥ 0.90	High
WV	2	1:1	Same	≥ 0.90	High
IN	9	1:8	Same	≥ 0.90	High
KY	6	1:5	Same	≥ 0.90	High
TN	9	1:8	Same	≥ 0.90	High
AL	7	3:4	Same	≥ 0.90	High
AR	4	1:3	Same	≥ 0.90	High
LA	6	1:5	Same	≥ 0.90	High
Growing metros (medium candidate CSS)					
WI	8	1:7	Same	0.75–0.85	Medium
MN	8	1:7	Same	0.75–0.85	Medium
IL	17	1:16	Same*	0.70–0.80	Medium
PA	17	1:16	Same*	0.70–0.80	Medium
VA	11	1:10	Likely	0.70–0.85	Medium
OR	6	1:5	Same	0.80–0.90	Medium
Sun Belt growth (low candidate CSS)					
AZ	9	1:8	Shifted	0.55–0.70	Low
NV	4	1:3	Shifted	0.50–0.65	Low
NC	14	6:8	Likely	0.65–0.80	Medium-low
CO	8	1:7	Shifted	0.60–0.75	Low-medium
GA	14	—	Unknown	—	Pending
TX	38	1:37	Shifted	0.50–0.65	Low
FL	28	1:27	Shifted	0.50–0.65	Low

* k changed between 2000 and 2020; normalised ratio comparison.

Table 4: Three-census CSS (2000/2010/2020). Package computation pending. Columns: seat stability (Boolean), ratio stability (Boolean), gap similarity ($[0,1]$), $\mathrm{CSS}_{3\mathrm{yr}}$ ($[0,1]$), stability class.

State	s_{seat}	s_{ratio}	s_{gap}	$\mathrm{CSS}_{3\mathrm{yr}}$	Class
IA	pending	pending	pending	pending	pending
KS	pending	pending	pending	pending	pending
WI	pending	pending	pending	pending	pending
NC	pending	pending	pending	pending	pending
AZ	pending	pending	pending	pending	pending
TX	pending	pending	pending	pending	pending
(full 50-state table pending package computation)					

Table 5: Proxy Jaccard stability for 30 comparable states (2000 vs. 2010). Proxy category based on $\Delta f = |f_{2000} - f_{2010}|$. High: $\Delta f < 0.05$. Medium: $0.05 \leq \Delta f < 0.15$. Low: $\Delta f \geq 0.15$. Full district Jaccard requires tract-level assignment files [computation needed].

State	f_{2000}	f_{2010}	Δf	Proxy Jaccard
High proxy Jaccard ($\Delta f < 0.05$)				
ME	0.50	0.50	0.00	High
WV	0.50	0.50	0.00	High
RI	0.50	0.50	0.00	High
MD	0.50	0.50	0.00	High
KS	0.25	0.25	0.00	High
NE	0.33	0.33	0.00	High
IN	0.11	0.11	0.00	High
KY	0.17	0.17	0.00	High
TN	0.11	0.11	0.00	High
AR	0.25	0.25	0.00	High
LA	0.17	0.17	0.00	High
WI	0.13	0.13	0.00	High
MN	0.13	0.13	0.00	High
OR	0.17	0.17	0.00	High
VA	0.09	0.09	0.00	High
IL	0.06	0.07	0.01	High
PA	0.06	0.07	0.01	High
OH	0.07	0.08	0.01	High
MI	0.08	0.08	0.00	High
CT	0.20	0.21	0.01	High
Medium proxy Jaccard ($0.05 \leq \Delta f < 0.15$)				
AL	0.46	0.40	0.06	Medium
NC	0.43	0.43	0.00	High
MS	0.25	0.30	0.05	Medium (boundary)
CO	0.13	0.13	0.00	High
NM	0.33	0.33	0.00	High
SC	0.43	0.36	0.07	Medium
NV	0.25	0.25	0.00	High
Low proxy Jaccard ($\Delta f \geq 0.15$)				
IA	0.18	0.49	0.31	Low
UT	0.32	0.25	0.07	Medium
AZ	0.11	0.11	0.00	High

Note: Values marked 0.00 for states with stable ratios are exact matches. *Entries without paired outputs are provisional and derived from 2020 p^*.*

Table 6: Stability count at different Δf thresholds (30 comparable states). A state is "stable" if $|f_{2000} - f_{2010}| \leq \tau$. The current paper uses $\tau = 0.05$ (bolded row).

Threshold τ	Stable states	Unstable states	Fraction stable
0.02	15	15	50 %
0.05	**20**	**10**	**67 %**
0.10	23	7	77 %
0.15	26	4	87 %

Table 7: Lorenz drift proxy: p^*_{2020} and predicted stability. $p^* = 0.5$ = equal split. $p^* \ll 0.5$ = asymmetric peel. States with $p^* < 0.1$ are most vulnerable to Lorenz drift.

State	k	p^*_{2020}	Predicted stability
Equal splits: Lorenz-stable			
IA	4	0.50	High (ratio-symmetric)
MD	8	0.50	High (ratio-symmetric)
MA	9	0.44	High (near-symmetric)
SC	7	0.43	High (near-symmetric)
AL	7	0.43	High (near-symmetric)
NJ	12	0.42	High (near-symmetric)
NC	14	0.43	Medium (near-equal; 6:8)
Moderate asymmetry: medium stability			
NE	3	0.33	Medium
NM	3	0.33	Medium
KS	4	0.25	Medium
NV	4	0.25	Low-medium (NV is growing)
CT	5	0.20	Medium
OR	6	0.17	Medium
KY	6	0.17	Medium
LA	6	0.17	Medium
Strong asymmetry: lower stability (Lorenz-drift-vulnerable)			
CO	8	0.13	Low-medium
WI	8	0.13	Medium (Milwaukee fixed)
MN	8	0.13	Medium (Minneapolis growing)
TN	9	0.11	Medium-high
AZ	9	0.11	Low (Phoenix growth)
IN	9	0.11	High (Indianapolis stable)
WA	10	0.10	Low-medium (Seattle growth)
VA	11	0.09	Medium
MI	13	0.08	Low-medium (Detroit decline)
OH	15	0.07	Low-medium
IL	17	0.06	Medium (Chicago fixed)
PA	17	0.06	Medium (Philadelphia fixed)

Table 8: 2×2 stability matrix: seed stability (B.7) $\times$ census stability (T.8 evidence and predictions). Representative states in each cell.

	Low census stability	High census stability
High seed stability	AZ, NV, CO, IA	KS, WI, IL, PA
Low seed stability	TX, FL, CA	(none expected)